

**Maternal exposure to perfluoroalkyl chemicals and anogenital distance in the offspring
A Faroese cohort study**

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1 **Title:**

2 Maternal exposure to perfluoroalkyl chemicals and anogenital distance in the offspring: a Faroese cohort
3 study

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30 **Abstract – 249 words**

31 Exposure to perfluoroalkyl substances (PFASs) has in some studies been associated with reduced anogenital
32 distance (AGD) in newborns as a sensitive indicator of prenatal anti-androgenic exposure. The aim of this
33 study was to investigate the association between maternal PFAS exposure and offspring AGD in a
34 population with wide ranges of PFAS exposures.

35
36 Participants were recruited in the Faroe Islands in 2007-2009, and information on AGD and PFAS exposure
37 was obtained from 463 mother-infant pairs. Perfluorooctanoic acid (PFOA), perfluorooctanesulfonic acid
38 (PFOS), perfluorohexane sulfonic acid (PFHxS), perfluorononanoic acid (PFNA) and perfluorodecanoic acid
39 (PFDA) were measured in maternal pregnancy serum. Data were analyzed using multiple linear regression
40 analysis adjusted for birth weight, child age at examination, parity, and maternal education level.

41
42 Among boys, higher maternal serum concentrations of PFOA, PFOS, PFNA and PFDA were significantly
43 associated with a longer AGD, both with the exposure entered as a continuous variable and as quartiles.
44 Boys in the highest quartile of PFOA, PFOS, PFNA and PFDA exposure had an increase in AGD of 1.2 mm
45 (95% CI 0.1;2.2), 1.3 mm (95% CI 0.3;2.3), 1.0 mm (95%, CI 0.0;2.0) and 1.3 mm (95%, CI 0.3;2.4),
46 respectively, when compared to boys in the lowest quartile of exposure ($p < 0.05$). No significant
47 association was found between male AGD and PFHxS. No association was found for girls.

48
49 In conclusion, elevated maternal exposure to major PFASs was significantly associated with a longer AGD in
50 boys. No significant associations were found among girls, thus suggesting a sex-dimorphic effect of PFAS
51 exposure.

52 **Keywords:** Perfluorinated compounds; PFAS; Prenatal exposure; Anogenital Distance; Faroe Islands

53

54

55 1. Introduction

56 Perfluoroalkyl substances (PFASs) are a group of highly persistent synthetically manufactured chemicals
57 used in fabrics and food packaging due to water-, stain-, and grease-resistant properties [1]. Human
58 exposure to PFASs occurs through ingestion of contaminated food and drinking water, inhalation of indoor
59 air and contact with other contaminated media [1-3]. Two of the previously most commonly used PFASs,
60 perfluorooctane sulfonic acid (PFOS) and perfluorooctanoic acid (PFOA), have been phased out by some of
61 the major manufacturers [2]. Nonetheless, other PFAS, such as perfluorohexane sulfonic acid (PFHxS),
62 perfluorononanoic acid (PFNA), and perfluorodecanoic acid (PFDA), are still in production, and these PFASs
63 have been studied to a lesser extent. PFASs cross the placental barrier and have been detected in human
64 amniotic fluid and umbilical cord blood [3].

65 Among other important adverse effects, PFASs have potential endocrine-disrupting effects
66 through affecting androgen receptor and estrogen receptor (ER) activity or altering the expression of
67 estrogen-responsive genes [4, 5] leading to hormonal imbalance, and may thus play a central role in
68 declining male reproductive health [6]. PFAS exposure has been associated with fetal growth, and a recent
69 review found that most studies report that maternal exposure to PFOA and PFOS was associated with low
70 birth weight [7, 8] but the mechanisms behind the influence of PFASs on fetal growth are largely unknown.
71 Animal studies have reported that exposure to PFASs could impair the reproductive system during the
72 developmental stage in male mice through an antiandrogen pathway[9, 10] and influence the expression of
73 estrogen-responsive genes in animal studies[11], and PFAS-induced changes in sex hormone biosynthesis
74 have been reported in vitro[12].

75 Anogenital distance (AGD), the distance from the anus to the genitalia, is routinely used in
76 animal toxicology studies and is sensitive to anti-androgenic exposure. In rodents, AGD is a sensitive
77 biomarker of androgen exposure during a critical embryonic window of testis development[13] and has
78 been shown to reflect the amount of androgen to which a male fetus is exposed in early development
79 where higher *in utero* androgen exposure results in a longer AGD [6, 14]. Two animal studies have found

80 that exposure to PFASs was associated with shorter AGD in male rat fetuses and wild male minks[15, 16].
81 The suggested mechanisms by which PFASs affects AGD is by increasing the ratio between estrogens and
82 androgens through transcriptional induction of the aromatase enzyme while also being a direct agonist to
83 the estrogen receptor (ER) [17-19]. Some of these findings have been reproduced in humans, e.g., in regard
84 to prenatal exposure to phthalates, which are anti-androgenic, and which have been found to reduce AGD
85 in boys [20]. Still, the potential effect on AGD is still poorly understood.

86 Three birth cohort studies have examined the association between maternal PFAS exposure
87 and AGD in the offspring, the results have been somewhat unclear. A Danish study found higher PFOS,
88 PFHxS, PFNA and PFDA concentrations to be associated with a shorter AGD in 3 month old girls, but a
89 longer AGD in boys, suggesting a sex-dimorphic effect [21]. A Chinese study found that high maternal PFOS
90 and PFDA exposure was associated with a shorter AGD in boys aged 1-3 days [22]. Lastly, a Canadian study
91 found an association between high PFOA and longer AGD in boys but not in girls [23]. Thus, the limited
92 evidence available shows somewhat unclear results and more studies are needed. We hypothesize that
93 maternal exposure to PFASs has impact on the development of the reproductive tract and AGD in the
94 offspring. Thus, the aim of this study was to investigate the association between maternal PFAS exposure
95 and AGD in infants in a Faroese birth cohort, where ranges of PFAS exposure are wide.

96

97 **2. Methods**

98 *2.1 Study population and data collection*

99 The study is based on data from the Faroese Birth Cohort 5, a population-based prospective cohort of 490
100 children, recruited between October 2007 and April 2009 with a participation rate of 73% [24-26]. Obstetric
101 information was obtained from midwife records and included maternal age, parity, pre-pregnancy weight,
102 height, education level, alcohol and smoking during pregnancy, partner smoking during pregnancy, and
103 birth weight.

104 Two weeks after expected term date, a pediatric examination was performed during which
105 AGD were measured. The AGD was measured with Vernier calipers according to a standardized method
106 described by Salazar-Martinez [27]. The infants were laid on their back with hips flexed and light pressure
107 was placed on the thighs until the hand of the examiner touched the subject's abdomen.. In boys, AGD was
108 determined as the distance from the center of the anus to the bottom of the scrotum (AGD_{AS}) and, in girls,
109 as the distance from the center of the anus to the posterior convergence of the fourchette (AGD_{AF}).
110 Measurement was repeated three times and a mean was calculated [27, 28]. All measurements were
111 performed by the same trained pediatrician. During the examination, i.e., two weeks after expected term
112 date, a sample of maternal blood was obtained for PFAS exposure assessment.

113 All protocols were approved by the Scientific Ethical Committee of the Faroe Islands and the
114 institutional review board at Harvard T.H. Chan School of Public Health. Written informed consent was
115 obtained from all mothers.

Table 1: Maternal characteristics (n=463) according to mean AGD (standard deviation) stratified by sex (n=232 males; n=231 females)						116
	N (%)	Males n=232		Females n=231		117
		%	AGD _{AS} mean (SD)	%	AGD _{AF} mean (SD)	118
All children	463		25.1(3.1)		13.4(2.1)	119
Maternal age (years)						120
<25	84 (18)	15	24.9 (2.8)	21	12.7 (1.8)*	
25-29.9	135 (29)	30	25.5 (2.8)	28	13.4 (1.8)	120
30-35	175 (38)	38	25.0 (3.1)	37	13.2 (2.4)	
>35	69 (15)	16	25.1 (3.5)	13	13.2 (1.8)	121
Parity						122
0	141 (31)	25	24.9 (2.9)	36	13.2 (1.8)	
1	160 (34)	37	25.0 (3.1)	33	13.7 (2.6)	122
2	102 (22)	26	25.5 (2.7)	18	13.0 (1.7)	
3+	59 (13)	12	25.6 (3.7)	13	13.6 (1.7)	123
Maternal pre-pregnancy BMI						124
<20	56 (12)	12	24.7 (2.8)	12	13.8 (2.1)	
20-25	248 (54)	52	24.9 (3.2)	55	13.3 (1.8)	124
>25	159 (34)	36	25.7 (2.9)	33	13.5(2.5)	
Gestational age at birth (weeks)						125
<38	31 (7)	7	25.4 (4.5)	7	13.7 (1.3)*	
38-40	302 (65)	58	25.2 (2.9)	72	13.2 (1.9)	126
>40	130 (28)	35	24.9 (3.0)	21	14.1 (2.6)	
Birth weight (g)						127
<2500	5 (1)	1	21.0 (.)*	2	14.3 (1.0)*	127
2500-4500	430(93)	92	25.1 (3.0)	93	13.3 (1.8)	
>4500	28 (6)	7	26.7 (2.4)	5	15.0 (4.7)	128
Weight at examination (g)						129
<3500	60 (13)	11	23.2 (2.9)*	15	12.4 (1.4)*	
3500-4500	300 (65)	60	24.7 (2.7)	69	13.5 (1.8)	129
>4500	103 (22)	29	26.7 (3.0)	16	14.1 (3.1)	
Age at examination (days)						130
<14	100 (22)	20	24.3 (2.3)*	23	12.9 (1.5)	
14-23	255 (55)	55	25.2 (3.2)	55	13.4 (2.3)	131
>23	108 (23)	25	25.8 (3.0)	22	13.7 (1.9)	
Maternal education level						132
Below high school	70 (16)	14	24.7 (2.6)	18	13.2 (2.1)	
High school or above	378 (84)	86	25.2 (3.1)	82	13.4 (2.1)	132
Maternal alcohol use during pregnancy						133
Yes	22 (5)	5	25.7 (3.4)	4	12.8 (1.3)	
No	441 (95)	95	25.1 (3.0)	96	13.4 (2.1)	133
Maternal smoking during pregnancy						134
Yes	73 (16)	15	25.2 (2.8)	17	13.4 (2.2)	
No	389 (84)	85	25.2 (3.1)	83	13.4 (2.1)	135
Partner smoking during pregnancy						136
Yes	152 (34)	33	25.0 (3.4)	34	13.3 (2.0)	
No	299 (66)	67	25.3 (2.8)	66	13.5 (2.2)	136
Note: : AGD _{AF} , anofourchette distance; AGD _{AS} , anoscrotal distance; ; SD, standard deviation						137
Missing values: parity: 1, maternal pre-pregnancy BMI: 1, maternal education level: 15, smoking during pregnancy: 1, partner smoking during pregnancy: 12						
*, marks a p-value <0.05 with one-way ANOVA test						

138 2.2

139

140

141 *Exposure assessment*

142 Maternal serum concentrations of PFASs (PFOA, PFOS, PFHxS, PFNA and PFDA) were measured by online
143 solid-phase extraction followed by high-pressure liquid chromatography with tandem mass spectrometry
144 [29, 30]. To ensure accuracy and reliability of data, each analytical series included quality control serum
145 samples, calibration standards, as well as reagent and serum blanks. Imprecision within and between
146 batches was less than 3.0 % and 5.2 %, respectively [31].

147

148 *2.3 Statistics*

149 Data were presented as counts and percentages for categorical variables, mean and
150 standard deviation (SD) or medians and interquartile range (IQR) for continuous variables. Pearson's
151 correlation coefficient was used to evaluate correlation between the five measured PFASs. Data was
152 assessed for normality by visual assessment using histograms and normal probability plots. One-way
153 ANOVA-test was used for normally distributed covariates to test for differences in AGD in relation to the
154 following covariates one at a time: maternal age, parity, maternal BMI, gestational age, birth weight,
155 maternal education level, maternal alcohol use during pregnancy, maternal smoking during pregnancy,
156 partner smoking during pregnancy, weight and age at examination adjusted to term-date. Differences in
157 PFAS concentrations between the same covariates were tested with Kruskal-Wallis' test for skewed
158 distributions. PFAS distributions were skewed and divided into quartiles, while also entered as a continuous
159 variable after logarithmic transformation to approach normality.

160 Stepwise multiple linear regression was used to examine the associations between PFASs
161 and AGD. Confounders included were factors known a priori to be important predictors of birth outcomes
162 or AGD [28, 29]. Weight at examination and infant's age at examination were included in all analyses. The
163 remaining covariates were gestational age (days), parity (0, 1, 2 and 3+), maternal smoking during
164 pregnancy (yes/no), partner smoking during pregnancy (yes/no), maternal alcohol use during pregnancy
165 (yes/no), maternal pre-pregnancy BMI (<20, 20–25, 25+ kg/m²), maternal education level (below high

166 school/high school or above). The only covariates that changed the beta estimate by at least 10 % were
167 parity and maternal education level, and both were therefore included in the final regression model.
168 Associations were reported in terms of beta estimates with 95% confidence intervals (95% CI), and
169 statistical significance was reached at a p-value of <0.05. All data analyses were performed using STATA/IC
170 16®.

171

172 **3. Results**

173 Of the cohort 490 mother-child pairs, a total of 463 pairs (94.4 % of the original cohort) had information on
174 both PFAS exposure and AGD available and were thus included in the analyses, i.e. 27 mother-child pairs
175 were excluded because of missing data. Mean maternal age at examination was 29.8 years and mean
176 maternal pre-pregnancy BMI was 24.3kg/m². Most women were multiparous (69 %) and of Scandinavian
177 origin (97 %). A total of 73 women (16 %) reported smoking and 22 (5 %) reported alcohol consumption
178 during pregnancy. The means of birth weight and AGD were 3787 g and 25.1 mm for boys and 3612 g and
179 13.4 mm for girls, respectively. For both sexes, birth weight, weight at examination, and age at examination
180 were significantly associated with the AGD while maternal age, gestational age at birth, and maternal
181 education level were significantly associated with AGD only among the girls (Table 1).

182 All measured PFASs were found in quantifiable concentrations in serum from the 463
183 women (Supplementary table). PFOS had the highest average concentrations (8.3 µg/L), followed by PFOA
184 (1.4 µg/L), PFNA (0.7 µg/L), PFDA (0.3 µg/L) and PFHxS (0.2 µg/L) and they were all mutually correlated
185 (Pearson's correlation coefficient r between 0.40 and 0.85, $p < 0.001$). Older women had significantly lower
186 concentrations of all five PFASs while women with higher parity had significantly lower concentrations of
187 PFOA, PFOS and PFHxS. Women with elevated PFOA exposure were more often smokers and gave birth to
188 children of lower birth weight and a lower weight at examination (Supplementary table). Significant
189 association was observed between exposure to all PFASs and birth weight (data not shown).

190 In multiple linear regressions, after adjusting for weight and age at examination (adjusted to
191 term date), parity and maternal education, higher maternal exposure to PFOA, PFOS, PFNA and PFDA was
192 significantly associated with a longer AGD_{AS} in boys, both with PFASs entered as continuous variable and as
193 quartiles (Table 2). Boys in the highest quartile of exposure to PFOA, PFOS, PFNA and PFDA had an AGD_{AS}
194 increase of 1.2 mm (95 % CI 0.1;2.2), 1.3 mm (95% CI 0.3;2.3), 1.0 (0.0;2.0) and 1.3 mm (95% CI 0.3;2.4),
195 respectively. No association was seen for PFHxS. No associations were found between maternal PFAS
196 exposure and AGD_{AF} in girls (Table 2).

Table 2: Linear regression analysis (β -coefficient and 95 % confidence intervals) on maternal pregnancy PFASs in quartiles, continuous (transformed by the use of natural logarithm) and AGD_{AS} and AGD_{AF}.

PFASs (ng/mL)	AGD _{AS} (mm) 232 males		AGD _{AF} (mm) 231 females	
	β	95 % CI	β	95 % CI
PFOA				
1 st	Reference		Reference	
2 nd	0.2	-0.8;1.1	0.8	-0.1;1.6
3 rd	1.2*	0.1;2.2	0.3	-0.6;1.2
4 th	1.1	0.0;2.3	0.2	-0.7;1.1
p-trend	0.02		0.85	
Continuous **	0.8*	0.1;1.5	0.0	-0.6;0.6
PFOS				
1 st	Reference		Reference	
2 nd	1.3*	0.3;2.4	0.1	-0.6;0.9
3 rd	1.3*	0.2;2.3	0.4	-0.5;1.1
4 th	1.3*	0.3;2.3	0.0	0.8;0.9
p-trend	0.02		0.81	
Continuous**	1.0*	0.1;1.8	0.1	-0.6;0.7
PFHxS				
1 st	Reference		Reference	
2 nd	-0.6	-1.6;0.4	0.3	-1.1;0.5
3 rd	0.1	-0.9;1.2	0.00	-0.8;0.8
4 th	0.5	-0.6;1.6	-0.5	-1.3;0.3
p-trend	0.18		0.39	
Continuous**	0.2	-0.3;0.7	-0.1	-0.4;0.3
PFNA				
1 st	Reference		Reference	
2 nd	0.2	-0.8;1.3	0.2	-0.7;1.0
3 rd	1.0*	0.0;2.0	0.1	-0.8;0.9
4 th	0.9	-0.2;1.9	-0.3	-1.1;0.4
p-trend	0.03		0.3	
Continuous**	1.1*	0.1;2.0	-0.1	-0.8;0.5
PFDA				
1 st	Reference		Reference	
2 nd	1.4*	0.4;2.5	0.3	-0.5;1.1
3 rd	1.0*	0.0;2.1	-0.1	-0.9;0.7
4 th	1.3*	0.3;2.4	-0.2	-1.0;0.6
p-trend	0.03		0.46	
Continuous**	0.8*	0.0;1.6	-0.3	-0.9;0.4

Note: Linear regression models are adjusted for parity, child age and weight at examination and mothers' level of education.
AGD_{AF}, anofourchette distance; AGD_{AS}, anoscrotal distance;;distance; PFASs, perfluoroalkyl substances Perfluorooctanoic acid (PFOA), perfluorooctanesulfonic acid (PFOS), perfluorohexane sulfonic acid (PFHxS), perfluorononanoic acid (PFNA) and perfluorodecanoic acid (PFDA).
Missing values: parity: 1; maternal education level: 15
*, marks a p-value <0.05
**, logarithmically transformed

4. Discussion

In this birth cohort study, maternal PFAS exposure was significantly associated with a longer AGD_{AS} in boys aged 14 days after term, whereas no association was found in girls.

To date, only three epidemiological studies have investigated the association between prenatal PFAS exposure and AGD, although showing somewhat uneven results (Table 3). Our findings of a longer AGD in boys exposed to PFAS is in agreement with a study from the Canadian Maternal-Infant Research on Environmental Chemicals Cohort (MIREC) that investigated the association between AGD_{AS}, anoscrotal distance, AGD_{AP}, anopenile distance, AGD_{AC}, anoclitral distance, and AGD_{AF}, anofourchette distance, and PFOA, PFOS, and PFHxS in 403 children [23]. They found that elevated PFOA was significantly associated to a longer AGD_{AS}, as was PFOS, albeit not significantly so. No association was observed with AGD_{AS} or among girls and PFASs. The Canadian study population resembled ours regarding maternal age and parity but had a higher level of education and with children at a lower average birth weight (Table 3). The PFOA concentration was similar, but the Canadian study measured PFAS in first trimester as opposed to a median of 19 days post-partum in our cohort. Interestingly, a Danish study (n=547) from the Odense Child Cohort (OCC) [21] also found increased AGD_{AS} in boys exposed to PFOS, PFNA and PFDA. Further, they reported a significant association between higher PFOS, PFHxS, PFNA and PFDA exposure and decreased AGD_{AC} in girls (Table 3). The Faroese and Danish studies are comparable regarding maternal characteristics and serum-PFAS concentrations although they measured PFAS in first trimester. The third study from the Shanghai-Minhang Birth Cohort, included 439 boys [22] and found significant associations between higher exposure to PFOS and PFDA and shorter AGD_{AS}, though not AGD_{AP} 1-3 days post-partum, i.e., findings

219 inconsistent with our results. However, the PFAS exposure in the Chinese boys was much higher compared
220 with our study (Table 3), and 84 % of the Chinese mothers were nulliparous which may have affected the
221 differences in results. Besides the three studies above, to our knowledge, only two animal studies have
222 investigated the association between PFAS exposure and AGD and with consistent associations. A reduction
223 AGD in male rat fetuses after exposure to PFOS was observed[15] and PFOS, PFDA, PFUA, and total PFASs
224 were related to shorter AGD in wild male minks[16]. Thus, our results are inconsistent with the findings in
225 animals.

226 In addition to AGD, a number of studies, both human and animal, have investigated the
227 effects of PFAAs on reproductivity and development[32]. In humans increases in exposure to PFOA and
228 PFOS has been associated with lower birth weight[7, 8], a finding supported by animal studies where
229 increased PFOA, PFOS, PFNA, and PFDA is associated with lower birthweight[1]. This is in accordance with
230 our study as we see that higher levels of all measured PFAS were significantly associated with lower
231 birthweight.

232

233

Table 3: A comparison of studies on PFAS exposure and AGD in infants

Study population	No of participants	PFAS concentrations in ng/mL	AGD measurements	Age at AGD-measurement	Confounders adjusted for	Results				
						PFOA	PFOS	PFHxS	PFNA	PFDA
Odense Child Cohort	316 boys 231 girls	Gestational age of 5-12 weeks:	AGD _{AS}	3.5 months (range 2.1–6.8 months)	Age at examination, weight for age Z-score, pre-pregnancy BMI, parity, smoking	NA	AGD _{AC} ↓*	AGD _{AC} ↓* ^Ω	AGD _{AC} ↓* ^Ω	AGD _{AC} ↓ ^Λ ^Ω
		PFOA 1.7	AGD _{AP}				AGD _{AS} ↑ ^Λ	AGD _{AS} ↓*	AGD _{AS} ↑*	AGD _{AS} ↑ ^Λ ^Ω
		PFOS 8.1	AGD _{AF}							
		PFHxS 0.3	AGD _{AC}							
		PFNA 0.7								
		PFDA 0.3								
Shanghai-Minhang Birth Cohort Study ¹	439 boys 0 girls	At gestational age of 12-16 weeks:	AGD _{AS}	1-3 days and 6 months	Age at examination, birth weight, parity, maternal education level, maternal age at delivery, gestational age, pre-pregnancy BMI	NA	AGD _{AS} ^{1-3d} ↓*	NA	NA	AGD _{AS} ^{1-3d} ↓*
		PFOA 20.1	AGD _{AP}				AGD _{AS} ^{6mo} ↓*			AGD _{AP} ^{1-3d} ↓*
		PFOS 10.7								
		PFHxS 2.8								
		PFNA 1.8								
		PFDA 2.1								
Maternal-Infant Research on Environmental Chemicals Cohort	198 boys 205 girls	At gestational age of <15 week:	AGD _{AS}	Mean 3.41 days	Varied, but could include recruitment site, education, gestational age, weight-for-length Z-score, active smoking status and household income	AGD _{AS} ↑*	NA	NA		
		PFOA 1.7	AGD _{AP}							
		PFOS 4.5	AGD _{AF}							
		PFHxS 1.1	AGD _{AC}							
Faroese birth cohort, Cohort 5	232 girls 231 boys	14 days after expected birth [§] :	AGD _{AS}	Mean 19.3 days	Age at examination, birth weight, parity, maternal education level	AGD _{AS} ↑* ^Λ ^Ω	AGD _{AS} ↑* ^Λ ^Ω	AGD _{AS} ↑ ^Λ	AGD _{AS} ↑* ^Λ ^Ω	AGD _{AS} ↑* ^Λ ^Ω
		PFOA 1.4	AGD _{AF}							
		PFOS 8.3								
		PFHxS 0.2								
		PFNA 0.7								
		PFDA 0.3								

Note: NA, no association; AGD_{AF}, anofourchette distance; AGD_{AS}, anoscrotal distance AGD_{AP}, anopenile distance; AGD_{AC}, anoclitoral distance; PFASs, perfluoroalkyl substances; Perfluorooctanoic acid (PFOA), perfluorooctanesulfonic acid (PFOS), perfluorohexane sulfonic acid (PFHxS), perfluorononanoic acid (PFNA) and perfluorodecanoic acid (PFDA)

1: Did not investigate separate quartiles

***: *p* <0.05 as continuous variable

Λ: *p* <0.05 in higher quartiles (q3 and/or q4)

Ω: *p*-trend <0.05

§ Mean 19.3 days after expected date of birth

234

235 The four epidemiological studies strongly indicate that PFASs exert a sex-dimorphic effect,

236 likely through endocrine disruption. This disruption of PFASs has also been studied in vitro and in animals.

237 Affected physiological pathways have been studied in vitro [11, 33-35]. One pathway is through PFOA and

238 PFOS decreasing testosterone levels by inducing aromatase activity through transcriptional activation of

239 CYP19, the aromatase enzyme and also induction of *CYP11B2*, coding for aldosterone synthase, which may

240 disrupt several of the functions related to aldosterone. Further, studies have found some PFASs to bind

241 directly to the ER with low affinity relative to endogenous estrogen [11, 33] and thereby exerting an
242 estrogenic effect and PFOA and PFOS are found to enhance the effect of estradiol on estrogen-responsive
243 genes [34]. In one animal study, blockage of androgen stimulation within a sensitive programming window
244 in fetal development resulted in de-masculinization of male genitalia including a shorter AGD, increased risk
245 of both hypospadias and cryptorchidism [14]. Generally, PFASs are suspected to cause endocrine disruption
246 through an increase in the fetal estrogen/androgen ratio. Following such disruption, one would expect a
247 shorter AGD in male offspring. . The suggested pathways which disrupt endocrine homeostasis found in
248 vitro and animal studies, support our findings of a sex-dimorphic effect of PFAS.

249 In humans, changes in the AGD relate to the symptom complex of diseases in male
250 reproductive system referred to as the testicular dysgenesis syndrome (TDS) [6], a syndrome causing
251 decreased fertility through a complex of symptoms such as decreased sperm quality, and an increased risk
252 of cryptorchidism and hypospadias [24, 36, 37]. The incidence of TDS is increasing and is hypothesized to be
253 linked to environmental pollutants, such as PFASs, that disrupt normal endocrine signaling, thereby causing
254 abnormal androgen action which alters the in utero development of the reproductive organs [38, 39].
255 Shorter AGD has been associated with, e.g., poorer semen quality [40, 41], genital malformations in men
256 [42, 43], and an increased risk of testicular germ cell tumor development [44]. Besides, longer AGD has
257 been suggested to be associated with higher sperm concentration, total sperm count, and total motile
258 sperm count in adult men and with fatherhood and may predict normal male reproductive potential [36,
259 45, 46]. In females, the long-term implications of AGD on reproductive health have only been studied
260 sporadically. In our study and the Canadian study [44], increased PFAS exposure was associated with longer
261 AGD, which may seem to contradict the TDS hypothesis, but cannot be considered as harmless, as the long-
262 term implications of longer AGDs are currently unknown.

263 Our study has several strengths. First, it is fairly large and comprised of a population-based
264 birth cohort with a high participation rate [24-26], and thus the cohort is likely to be representative of the
265 general population, thereby limiting selection bias. Further, the birth outcomes were objectively reported

and AGD was measured three times by the same trained pediatrician to eliminate inter-examiner variance and reduce imprecision. However, some weaknesses need consideration. PFASs were measured approximately two weeks post-partum, i.e., several months after the hypothesized masculinization programming window occurring between 8- and 14 weeks of gestation [14]. Exposure misclassification may have occurred, as maternal serum concentrations of PFOS, PFOA, and PFNA average higher in the first trimester, as compared to the second and third trimester [47, 48]. However, due to the long half-life of PFASs, one can assume that PFAS concentrations, measured postpartum after the initiation of breastfeeding, were likely comparable or even lower than PFAS concentrations during the relevant stage of development [37]. The Faroese women were exposed to lower PFAS concentrations than the Chinese women, whereas the exposure levels were relatively similar to Danish and Canadian pregnant women (table 3). Additionally, unknown confounders associated with PFAS exposure and intra-uterine growth may exist. e.g. lifestyle, and exposure to other chemicals, including PCBs and methylmercury, that the Faroese population are highly exposed to [49]. Lastly, the number of endocrine disruption chemicals (EDCs) that the mothers have been exposed to is not limited to the five PFASs measured, and serum-PFAS concentrations could conceivably function as a surrogate for other EDCs that may be highly correlated. Of note, studies exist that associate other EDCs with shorter AGD, e.g. phthalates [6], which is opposite to the findings of the present study.

283

284 **5. Conclusions**

285 In conclusion, we found that maternal PFAS exposure was significantly associated with a longer AGDs in
286 boys. No significant associations were found among girls, suggesting sex-dimorphic effects of PFASs.
287 Whether the observed association with longer ASD and PFAS influences the reproductive health of males is
288 unknown and needs to be investigated further.

289

290

291 **Conflict of interest**

292 The authors declare that they have no competing financial interests.

293

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297

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Supplementary table: Maternal characteristics (n=463) according to median (inter quartile range) maternal serum concentrations of PFASs (ng/ml)

	N (%)	PFOA	PFOS	PFHxS	PFNA	PFDA
All women	463	1.4 (1.0-2.0)	8.3 (6.2-11.7)	0.2 (0.1-0.3)	0.7 (0.5-0.9)	0.3 (0.2-0.4)
Maternal age (years)						
<25	84 (18)	1.7 (1.4-2.3)*	8.0 (6.2-10.0)*	0.2 (0.1-0.3)*	0.6 (0.5-0.8)*	0.2 (0.1-0.3)*
25-29.9	135 (29)	1.4 (1.0-2.0)	8.7 (6.6-10.7)	0.2 (0.1-0.3)	0.6 (0.5-0.8)	0.3 (0.2-0.3)
30-35	175 (38)	1.1 (0.9-1.8)	8.0 (5.6-10.4)	0.2 (0.1-0.3)	0.7 (0.5-0.9)	0.3 (0.2-0.4)
>35	69 (15)	1.2 (0.9-1.8)	9.4 (6.3-12.8)	0.3 (0.2-0.4)	0.8 (0.6-1.0)	0.3 (0.2-0.4)
Parity						
0	141 (31)	2.0 (1.5-2.5)*	8.6 (7.0-10.8)*	0.2 (0.2-0.4)*	0.7 (0.5-0.9)	0.2 (0.2-0.3)
1	160 (34)	1.3 (1.0-1.7)	8.6 (6.3-10.6)	0.2 (0.1-0.3)	0.6 (0.5-0.8)	0.2 (0.2-0.3)
2	102 (22)	1.1 (0.8-1.6)	8.0 (5.2-10.8)	0.2 (0.1-0.3)	0.7 (0.5-0.9)	0.3 (0.2-0.4)
3+	59 (13)	0.9 (0.7-1.1)	7.0 (5.1-9.6)	0.1 (0.1-0.3)	0.7 (0.5-0.8)	0.3 (0.2-0.4)
Maternal pre-pregnancy BMI						
<20	56 (12)	1.4 (0.9-2.0)	7.8 (5.8-10.2)	0.2 (0.1-0.3)	0.7 (0.5-0.8)	0.3 (0.2-3.3)
20-25	248 (54)	1.3 (0.9-2.0)	8.1 (6.0-10.6)	0.2 (0.1-0.3)	0.6 (0.5-0.9)	0.3 (0.2-0.4)
>25	159 (34)	1.4 (1.0-1.9)	8.7 (6.7-11.2)	0.2 (0.1-0.3)	0.7 (0.5-0.9)	0.2 (0.2-0.4)
Gestational age at birth (weeks)						
<38	31 (7)	1.7 (0.9-1.9)	9.0 (6.3-11.1)	0.3 (0.2-0.3)	0.7 (0.6-0.8)*	0.3 (0.2-0.3)
38-40	302 (65)	1.4 (0.9-2.0)	8.9 (6.3-10.5)	0.3 (0.1-0.3)	0.8 (0.5-0.9)	0.3 (0.2-0.4)
>40	130 (28)	1.5 (1.0-1.8)	8.5 (5.3-11.1)	0.2 (0.1-0.3)	0.7 (0.5-0.8)	0.3 (0.2-0.3)
Birth weight (g)						
<2500	5 (1)	1.8 (1.8-2.0)*	9.0 (6.9-11.5)	0.2 (0.2-0.3)	0.7 (0.7-0.7)	0.3 (0.3-0.4)
2500-4500	430 (93)	1.4 (1.0-2.0)	8.4 (6.2-10.8)	0.2 (0.1-0.3)	0.7 (0.5-0.9)	0.3 (0.2-0.4)
>4500	28 (6)	0.9 (0.7-1.4)	7.6 (5.4-9.3)	0.2 (0.1-0.3)	0.6 (0.5-0.8)	0.3 (0.2-0.3)
Weight at examination (g)						
<3500	60 (13)	1.6 (1.0-2.4)*	8.9 (7.0-11.0)	0.3 (0.2-0.4)*	0.7 (0.6-1.0)*	0.3 (0.2-0.4)
3500-4500	300 (65)	1.4 (1.0-2.0)	8.3 (6.2-10.6)	0.2 (0.1-0.3)	0.7 (0.5-0.9)	0.3 (0.2-0.3)
>4500	103 (22)	1.1 (0.8-1.6)	7.7 (5.6-9.9)	0.2 (0.1-0.3)	0.6 (0.5-0.8)	0.3 (0.2-0.3)
Age at examination (days)						
<14	100 (22)	1.7 (1.0-2.4)	8.9 (7.1-11.1)	0.3 (0.2-0.4)	0.7 (0.6-1.0)	0.3 (0.2-0.4)
14-23	255 (55)	1.4 (1.0-1.9)	8.2 (6.2-10.6)	0.2 (0.1-0.3)	0.7 (0.5-0.9)	0.3 (0.2-0.3)
>23	108 (23)	1.1 (0.8-1.6)	7.7 (5.6-9.9)	0.2 (0.1-0.3)	0.6 (0.5-0.8)	0.3 (0.2-0.3)
Maternal level of education						
Below high school	70 (16)	1.6 (1.1-2.2)*	8.2 (6.2-11.0)	0.2 (0.1-0.3)	0.7 (0.5-0.9)	0.2 (0.2-0.3)
High school or above	378 (84)	1.3 (0.9-1.9)	8.2 (6.2-10.5)	0.2 (0.1-0.3)	0.7 (0.5-0.9)	0.3 (0.2-0.4)
Maternal alcohol use during pregnancy						
Yes	22 (5)	1.5 (0.9-1.7)	8.4 (6.8-11.2)	0.2 (0.1-0.3)	0.6 (0.5-0.9)	0.3 (0.2-0.4)
No	441 (95)	1.4 (1.0-2.0)	8.3 (6.2-10.6)	0.2 (0.1-0.3)	0.7 (0.5-0.9)	0.3 (0.2-0.4)
Maternal smoking during pregnancy						
Yes	73 (16)	1.6 (1.1-2.0)*	8.8 (6.8-10.6)	0.2 (0.2-0.3)*	0.6 (0.5-0.9)	0.3 (0.2-0.4)
No	389 (84)	1.3 (0.9-1.9)	8.2 (6.1-10.6)	0.2 (0.1-0.3)	0.7 (0.5-0.9)	0.3 (0.2-0.4)
Smoking partner during pregnancy						
Yes	152 (34)	1.5 (1.1-2.1)*	8.7 (6.2-11.2)	0.2 (0.1-0.3)	0.7 (0.5-0.9)	0.3 (0.2-0.3)
No	299 (66)	1.3 (0.9-1.9)	8.2 (6.2-10.5)	0.2 (0.1-0.3)	0.7 (0.5-0.9)	0.3 (0.2-0.4)

Note: AGD, anogenital distance; PFASs, perfluoroalkyl substances; Perfluorooctanoic acid (PFOA), perfluorooctanesulfonic acid (PFOS), perfluorohexane sulfonic acid (PFHxS), perfluorononanoic acid (PFNA) and perfluorodecanoic acid (PFDA)*p<0.05, Kruskal Wallis' test.

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